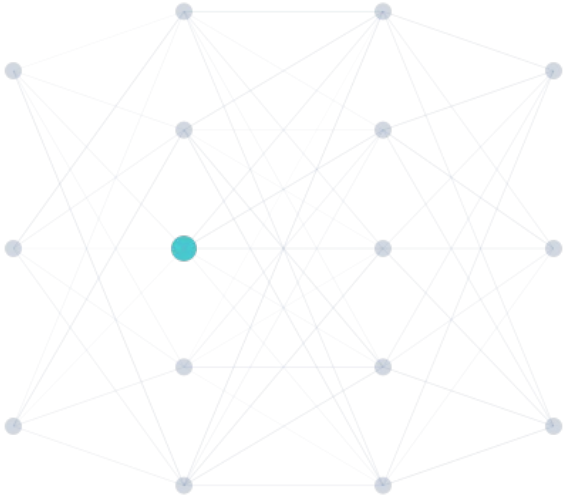
THE PROBLEM

# AI is scaling.
# Not getting smarter.



Billions in compute. Marginal gains in robustness.

The brain is a physical representation of energy-efficent compute.

*Something fundamental is missing.*

THE THESIS

# Evolution already solved this.

We're learning to read the solution.

500 million years of neural optimization

— available as a training signal.

### Biology encodes geometry

Neural firing patterns form a structured similarity space over inputs.

### Geometry is distillable

The signal can be extracted and synthesized.

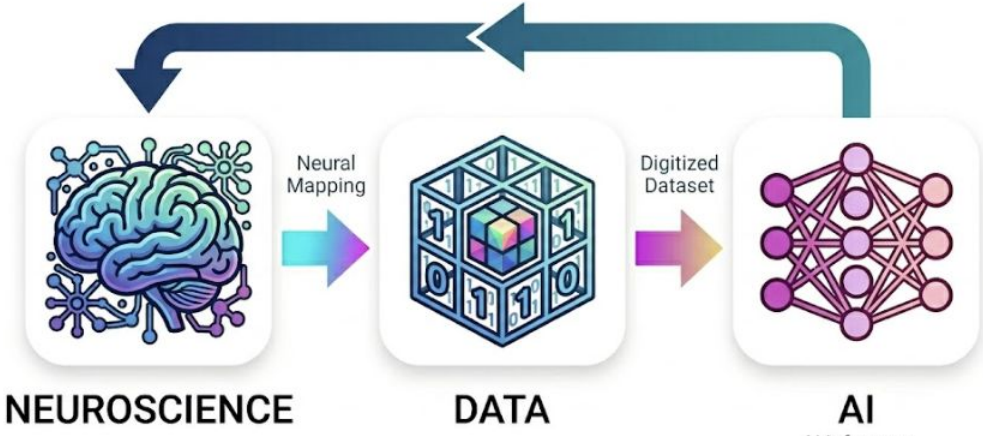
### Priors ship in any model

Injected at training time. Zero biological overhead at deployment.

THE THESIS

# We extract recipes from real neurons

**brain compute is ~million times more energy-efficent!*




**Three products:**

1. Neuronal Datasets → improve AI accuracy (**this demo**)
2. Inference Algorithms → energy-efficient AI (**patent filled Nov 2025**)
3. Bio-compute→ neurons on silicon for ultra low energy AI training

What AI interpretability researchers try to infer, we can measure directly.
De-risking ideation and fastening design search for AI.

THE PROOF

# Validation: biocompute as a data factory

The whole bio-computing field is relying on the **assumption** that neuronal activity is unique and will bring improvements to compute and AI.

**We shared that hope too!**
But, instead investing first in scaling the biocomputing technology (that need further resolution improvement) - we conducted this proof.
As trained neuroscientist we knew how to use a limited neuroscience data for a small demo and distill the important facts.

We conclude that there is **potential for improving current AI models accuracy** when using neuronal activity.
Therefore:

- building biocomputing data factory is marketable
- our data sets can be widely adopted across any type of AI models, including multi-modal AI and world models. That is currently a 3,000+ companies world-wide, rapidly growing.

**Revenue opportunity: $0.2-1M/customer/year**

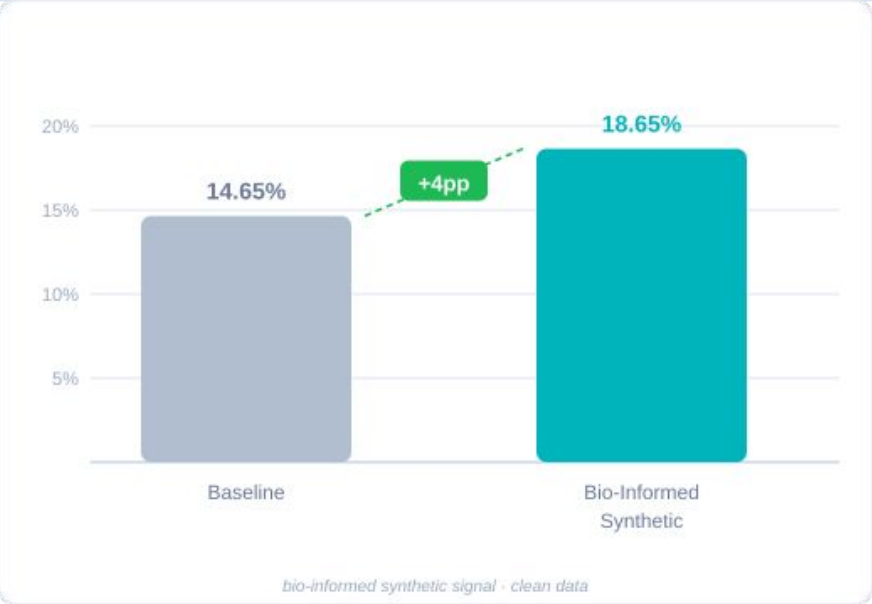
***further research needed, we are on it.**

FINDING

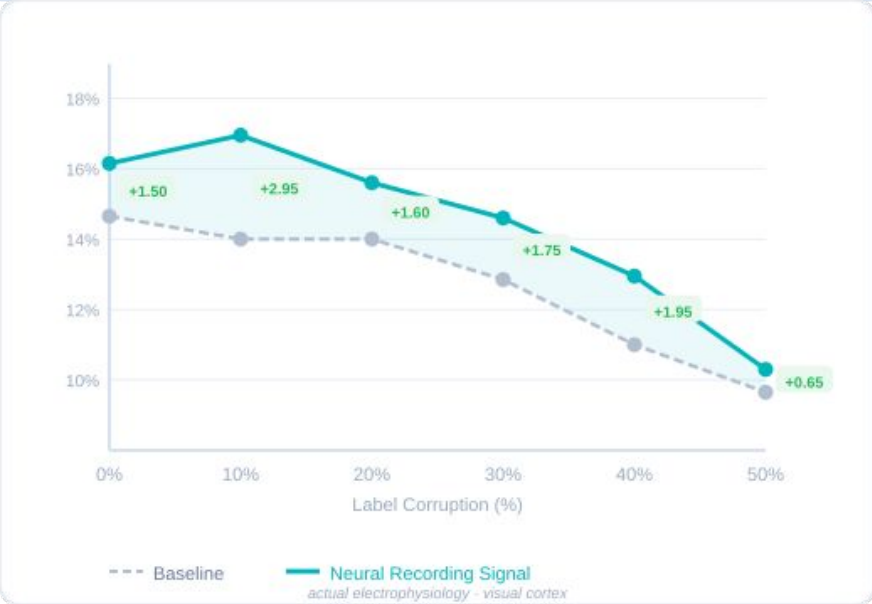
# Neural geometry improves AI vision.

*Two signals. Same conclusion: the lift is real and robust.*

**Best accuracy lift · bio-informed synthetic signal**




**Robustness · neural recording signal · 0–50% label noise**



HOW WE TESTED IT

# High-level.

*Proof-of-concept scale — directional finding confirmed.*

CIFAR-100 subset (10%)— 5,000 training images

We got a +4pp lift on 10% of the training data. **Imagine what happens at full scale.**

**Neural Signal**
**Electrophysiology**
Visual cortex recordings
converted to firing rates
over natural scene images

**Visual Benchmark**
**Standard image dataset**
Public image recognition
benchmark — same task
used across the field

**Model**
**Small CNN**
Proof-of-concept scale
3-layer convolutional net
directional, not optimized

***further research needed, we are on it. More detail available for investors on demand**

Foundation Team

# Cross-disciplinary talent. Ready day one.

**Mohsen Sharifi-Renani, PhD**
Head of research,
*Ex-Spotify, Ex-Optum*

- Builds LLMs from scratch
- Ships AI at scale
- Role: ML/AI research and engineering

**Tomasz Gorka, MBA**
Head of product,
*Ex-Google, Ex-CTO*

- Distributed systems, agentic AI
- Founded startup, led unicorn eng team
- Role: building AI native company infrastructure, AI engineering

**Marta Gajowa, PhD**
CEO, Neuroscientist, *Ex-UC Berkeley*

- 15+ years building neurotechnology serving neuronal circuits reverse-engineering
- Multiple publications (Nat. Neurosci., Neuron, NeurIPS)
- Invited SfN 2025 speaker, multiple competitive awards, SF LLM day 2026 speaker
- Role: vision execution, research, engineering

**Our Competitive Advantages:** ✓ Proprietary data ✓ Neurotechnology ✓ **Patent filed** ✓ Cross-disciplinary ✓ Early hires committed ✓ Truly AI-native company ✓ **3 LOI signed pre-product**

THE COMPANY

# Neuraffica is fundraising now

*Distilling 500M years of evolution into AI.*

**Read** Measure neural activity at single-neuron resolution at scale.

**Engineer** Design and reshape connectivity at circuit level

**Distill** Extract computational principles. Package them as portable energy-efficient AI solutions.

**The Ask: $5M pre-seed round**

**Use of funds:**

- Scale data factory
- Algorithmic models
- Close first enterprise deals
- Launch bio-compute in-house pilots
- Team full execution

**Neuraffica: Running AI like neurons, on neurons. We will be first, not just the best!**